QTL Mapping – User Guide

Welcome to our QTL Mapping and Genetic Analysis Tool!
This system enables researchers to analyse genotype and trait data using advanced QTL detection techniques like Single Marker Test (SMT) and Single Interval Mapping (SIM). The program is user-friendly and offers both a GUI interface and a script-based command-line alternative.

Installation & Setup:

This project is built using **Python 3.11.9**.

It is recommended to use **Python 3.11.9 or later** for full compatibility.

Open bash and execute the following steps:

1) **Clone the Project:**
   run:
   git clone https://github.com/Tamert98/QTL-Mapping-Project.git

2) **(Recommended) Create a Virtual Environment:**
   To avoid dependency conflicts, it's recommended to use a virtual environment. Run the following commands to create and activate a virtual environment:
   run:

   python -m venv qtl-env
   source qtl-env/bin/activate

3) **Install Required Packages:**
   This will install all the dependencies specified in the requirements.txt file, including: numpy, matplotlib, scipy ,Pillow ,customtkinter
   run:

   pip install -r requirements.txt

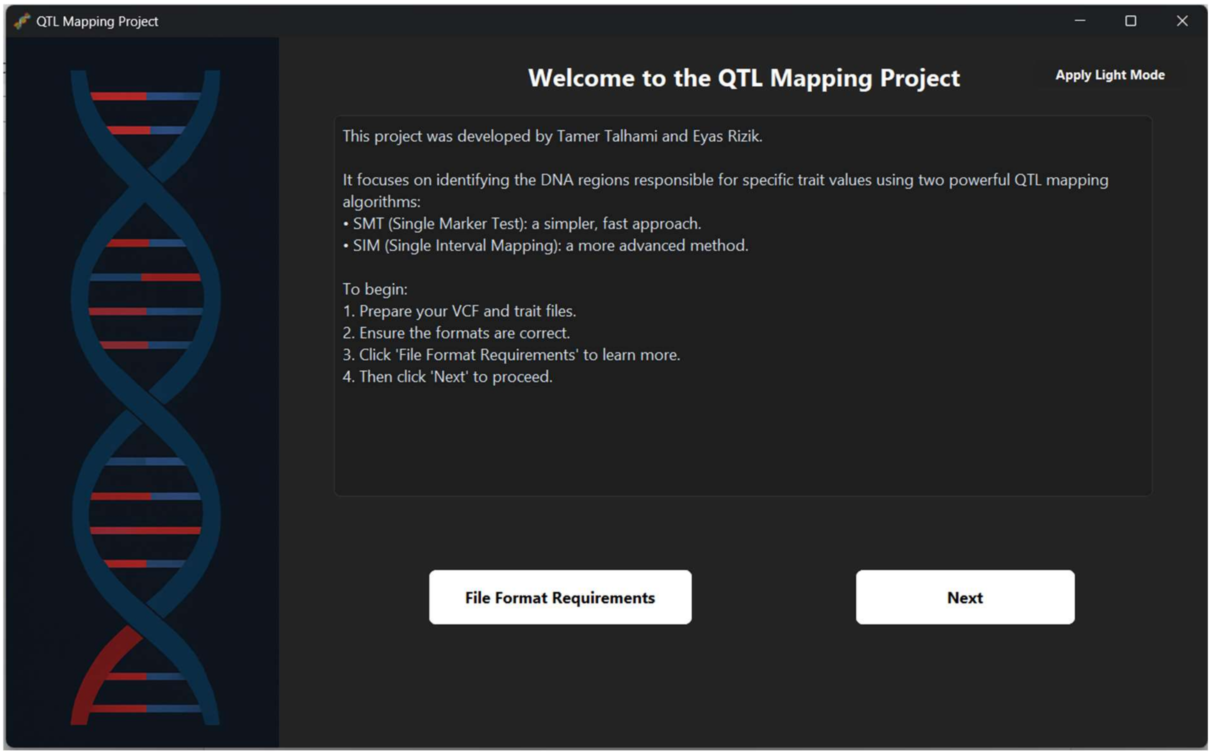
Running the app :

After installing the project and all required dependencies, you can navigate to main_app.py and run the program.

How to use it:

For a visual walkthrough, we've included a **demonstration video** in the GitHub repository under the folder Final Project - Phase B.
We highly recommend watching it to understand how to use the system effectively.

1) Front Page Overview :



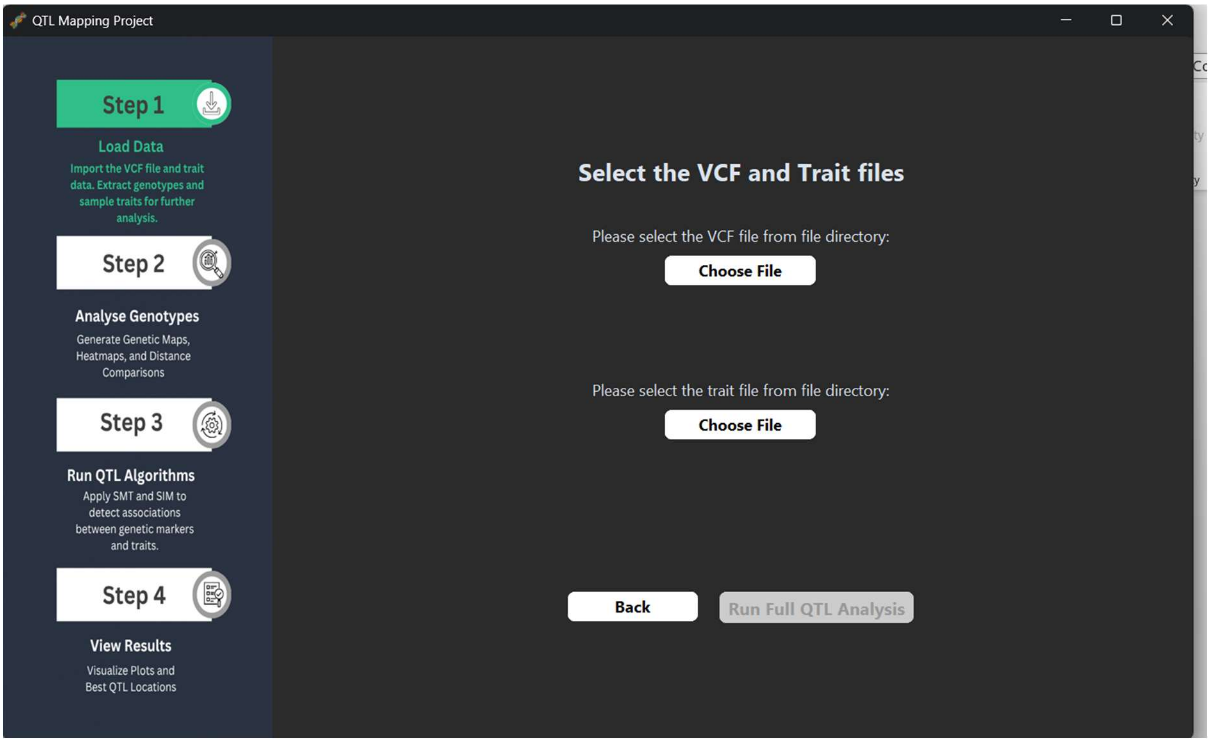
This is the **main screen** of the application.

- It provides a clear introduction to the project, including:
    - The goal of identifying trait-associated DNA regions using **SMT** and **SIM** QTL mapping algorithms.
    - Simple instructions on how to get started.
- **Instructions for required input files** (VCF and trait file) are briefly mentioned here.
  For more detailed file format guidance, click the **"File Format Requirements"** button.
- A **Dark/Light Mode toggle** is available in the top-right corner.
  Simply click **"Apply Dark/Light Mode"** to switch themes based on your preference.

This page ensures that users understand what's needed before beginning the analysis.

Once you have your folders prepared, select **"Next"** to go to the next step.

2) Select VCF and Trait Files Page:



On this page, you are required to load the **VCF** and **Trait** files.

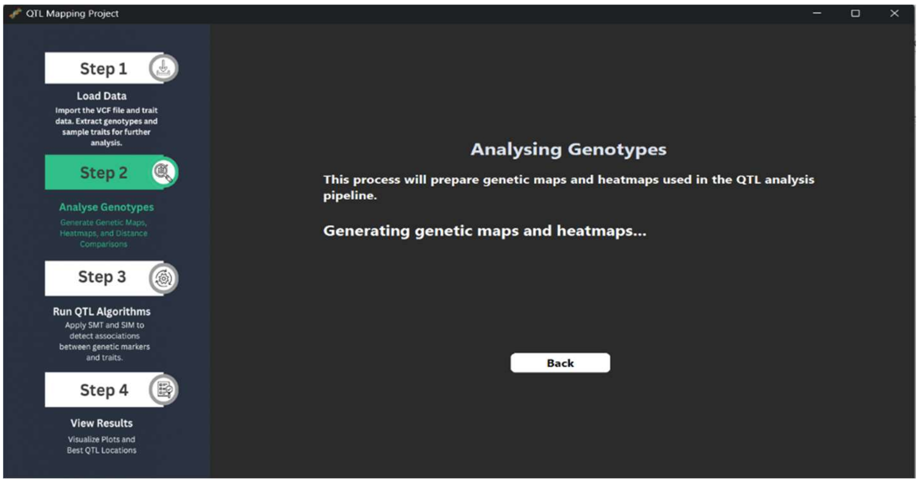
- To do this, click the **“Choose File”** button for each file type.
- A file selection window will open.
  We enforce that the files end with .vcf for the VCF file and .txt for the trait file.

Once both files are selected:

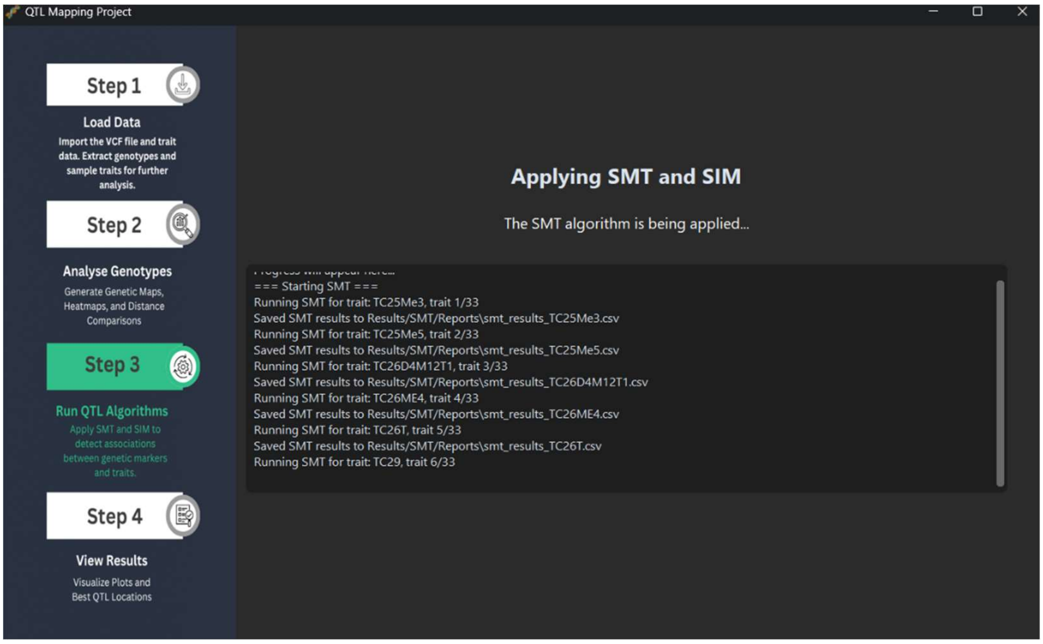
- The **"Run Full QTL Analysis"** button will turn **green** and become **enabled**.
- Click it to begin the full QTL mapping pipeline.

3) QTL analysis run page:

At this stage, the program performs the full analysis process. It begins with processing the genotype data,



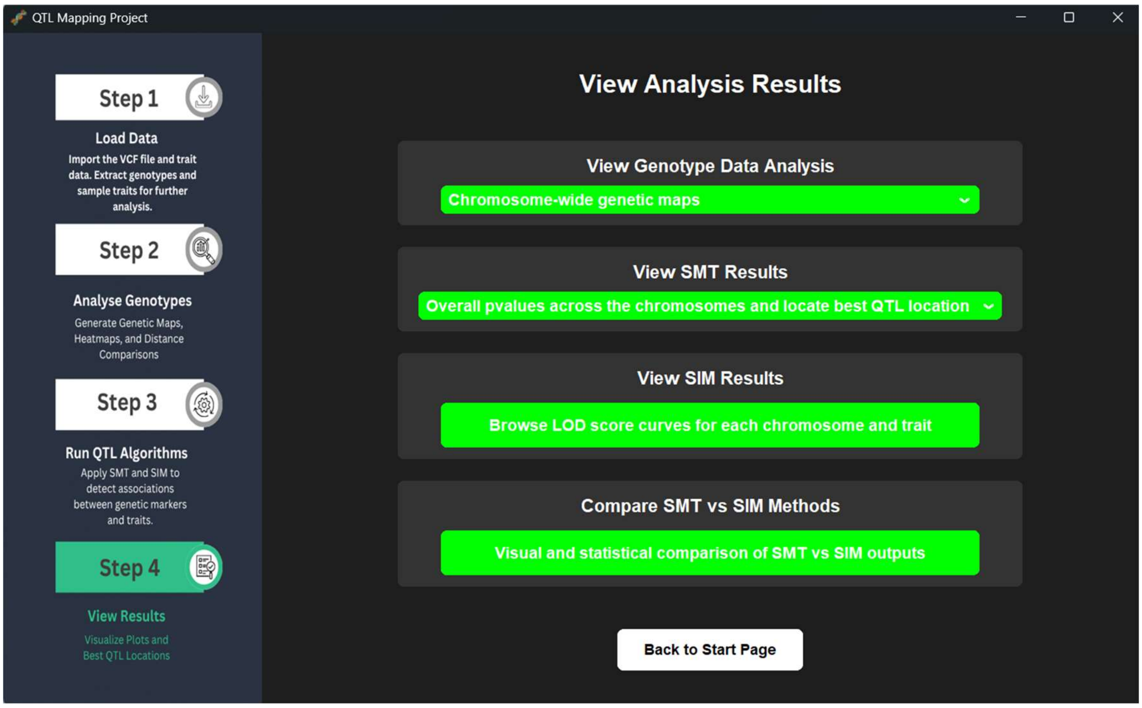
followed by running the SMT and SIM algorithms.



This step may take some time depending on the size of your dataset.
For example, in our case, with 100 individuals, 30 traits26 chromosomes, each with around 700 markers.
The full analysis took approximately 25 minutes.
Once the computation is complete, the program will automatically proceed to the Results Viewer.

4) View Results page:

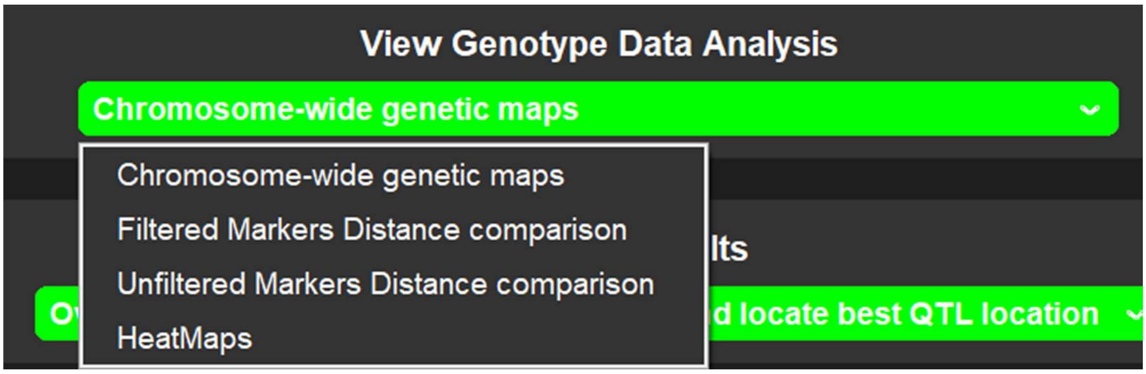
This is the final stage of the pipeline. Here, you can explore all results generated during the QTL analysis. The interface includes four major sections:



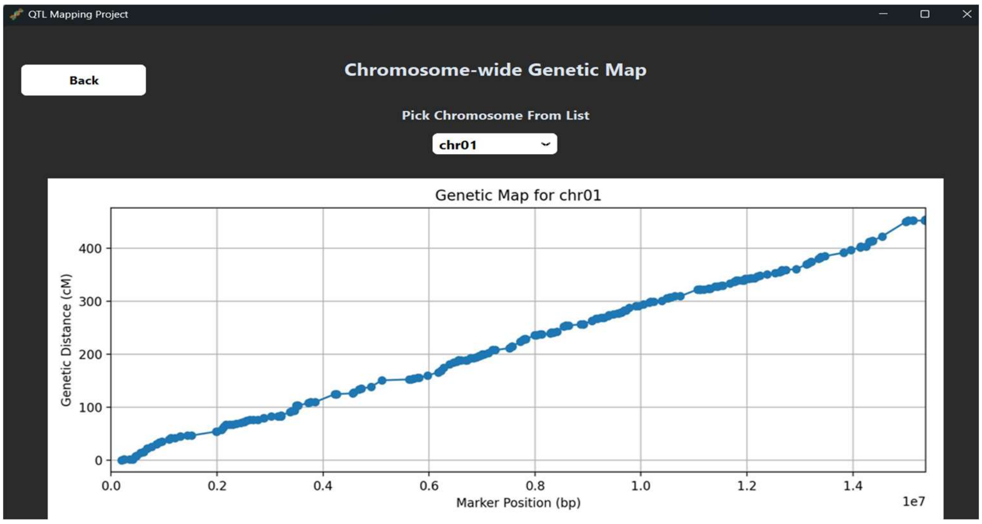
A) **Genotype Data Analysis**

This section offers four visualization options:

- **Chromosome-wide genetic maps**
- **Filtered markers distance comparison**
- **Unfiltered markers distance comparison**
- **Heatmaps**



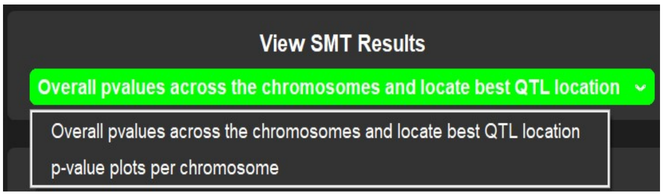
You can select any of these options from the dropdown.
For example, choosing "Chromosome-wide genetic maps" will lead to a page like the one shown, where you can select a specific chromosome and view its genetic map.



After pressing back you will go back to the View results main page

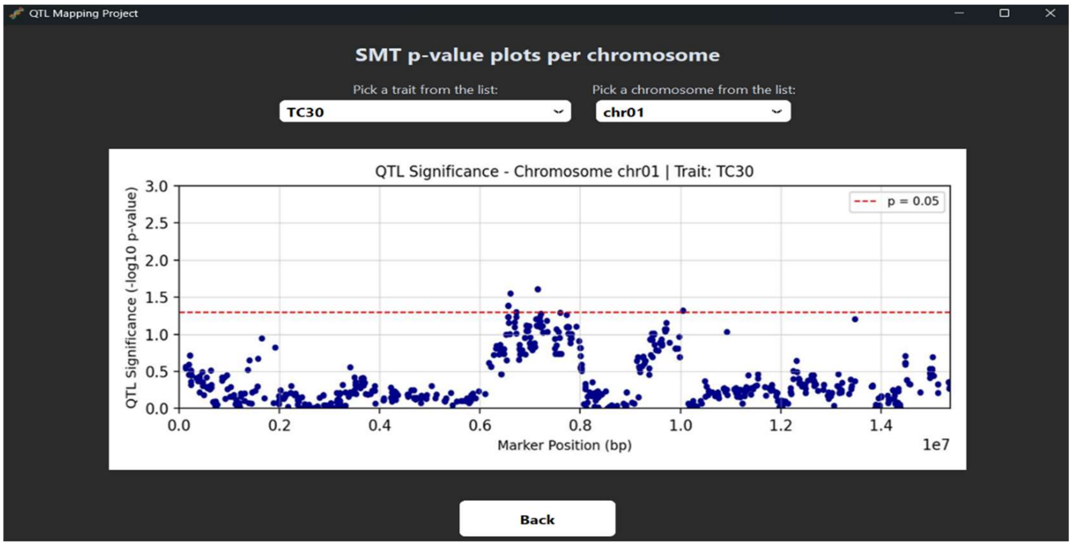
B) SMT Results :

Now moving to the SMT section — you can select between two visualization options:



- Overall p-values across all chromosomes
- P-value plots per chromosome

For example, if you choose p-value plots per chromosome, you will be redirected to a new screen. There, you can select both the trait and the chromosome from dropdown menus to display the corresponding p-value plot.



C) SIM Results :

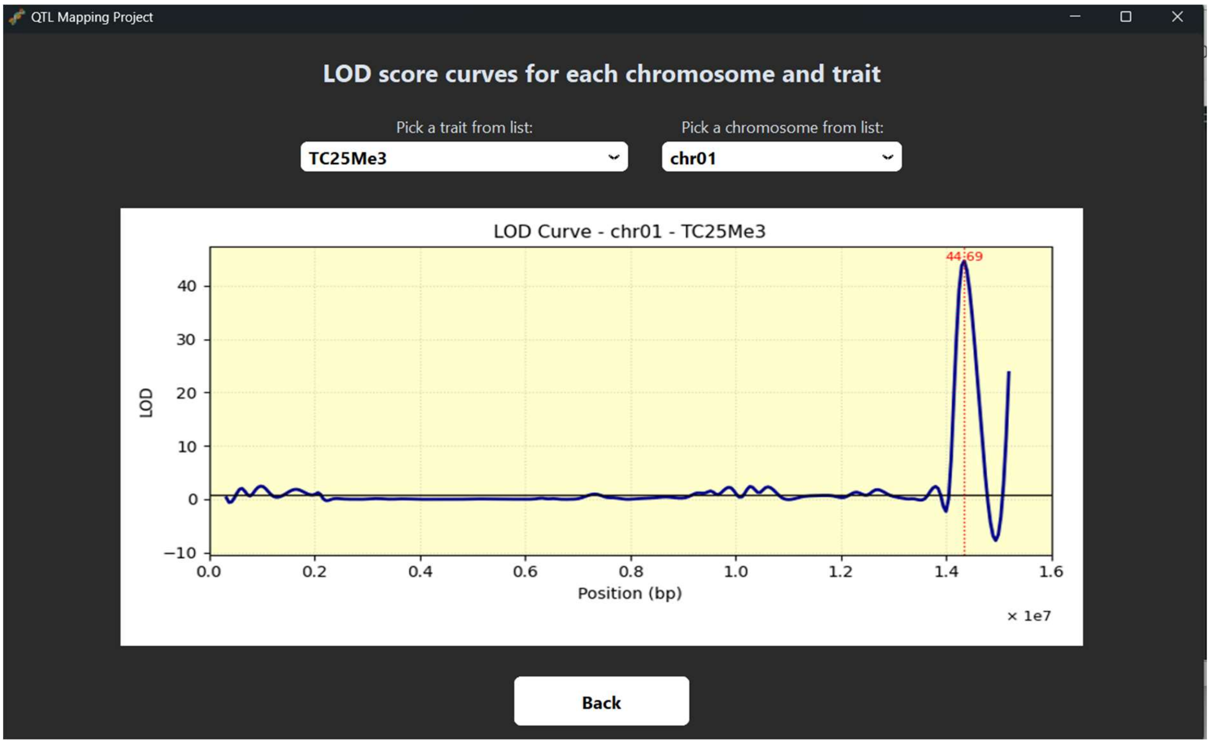
Now moving to the SIM Results section:

This screen displays the LOD score curve for a selected trait and chromosome. You can choose the trait and chromosome from the dropdown menus at the top.

The graph shows:

- The LOD score along the chromosome.
- The peak LOD value, indicating the most likely QTL position.
- A red vertical line marking the estimated QTL location with the highest likelihood.

This visualization helps assess how strongly each genomic position is associated with the trait, allowing for precise localization of potential QTLs.



D) Comparison of SMT vs SIM Outputs:
In this final section, you can visually compare the results of the SMT and SIM algorithms side by side.

You can:

- Select a trait and a chromosome from the dropdown menus.
- View the SMT p-value plot on the left.
- View the SIM LOD score curve on the right.

This comparison helps you evaluate:

- Whether both methods point to the same QTL region.
- The strength and clarity of the signal using each algorithm.
- Cases where one method may detect associations missed by the other.

This dual-panel view is useful for deeper interpretation and validation of candidate QTLs.